

Lecture 12 Worksheet

Math 124, Fall 2026 at the University of Michigan

Some lectures will have an accompanying worksheet. We will do our best to include these problems directly in the notes as activities or worked examples.

Activity 1

A **linear transformation** $T : \mathbb{R}^d \rightarrow \mathbb{R}^n$ is a function with domain $\mathbb{R}^d$ and codomain $\mathbb{R}^n$.

- a) State the two properties that make T a linear transformation.

- b) Suppose T is a linear transformation, $T(\vec{u}) = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, and $T(\vec{v}) = \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix}$. Find $T(2\vec{u} - \vec{v})$ without finding a matrix for T .

- c) Explain why every linear transformation sends $\vec{0}$ to $\vec{0}$. Use this to show that $F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x+2 \\ y \end{bmatrix}$ is not a linear transformation.

- d) The function $G\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} xy \\ y \end{bmatrix}$ sends $\vec{0}$ to $\vec{0}$. Is it a linear transformation? Give a specific counterexample or explanation.

Activity 2: From a function to a matrix

Consider the function

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} x + y \\ x \end{bmatrix}.$$

In words, add the two input entries to get the first output entry, and copy the first input entry to get the second.

- a) Find $T \left(\begin{bmatrix} 2 \\ -1 \end{bmatrix} \right)$, $T \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$, and $T \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$.

- b) Find a matrix A such that $T(\vec{v}) = A\vec{v}$ for every input $\vec{v}$. Check your answer by multiplying $A \begin{bmatrix} x \\ y \end{bmatrix}$.

- c) Now consider $U : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, where

$$U \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} x \\ y \\ x + y \end{bmatrix}.$$

Find its matrix. Explain how its shape matches its domain and codomain.

- d) Can U produce the output $\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$? What about $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$? Explain using the rule for U .

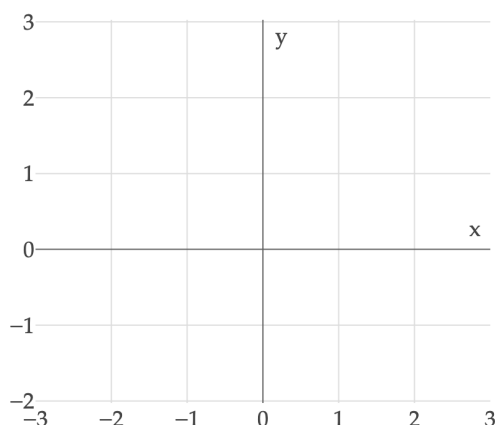
Activity 3: What happens to a square?

Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. We'll use a square to see what multiplying by A does.

- a) Compute the outputs for the four corners of the unit square:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

- b) Draw the original square on the left. On the right, plot its four outputs and connect them in the same order. What shape do you get?



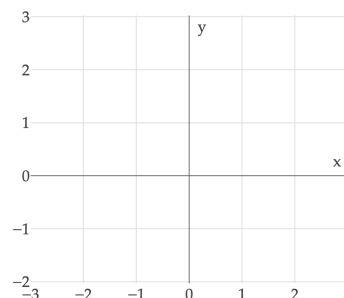
Before



After multiplying by A

- c) A different transformation rotates vectors clockwise by 90° about the origin. Where do $\vec{e}_1$ and $\vec{e}_2$ go? Use those two outputs to find its matrix R .

- d) Use your matrix R to rotate $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Sketch the input and output as arrows from the origin on the grid to the right.



Activity 4: Describe it, then find its matrix

For each linear transformation below, first find $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$ for any x and y . Then, find the corresponding matrix. Finally, use the matrix to find $T\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right)$.

- a) Stretch horizontally by a factor of 2, leaving the vertical coordinate alone.

- b) Reflect across the x -axis. Which coordinate changes sign?

- c) Add the vertical coordinate to the horizontal coordinate, and leave the vertical coordinate alone. Which points stay fixed?

- d) Project orthogonally onto the x -axis. What happens to the second coordinate? Give two different input vectors that produce the same output.

Activity 5: Matrix multiplication

Golden rule: The number of columns in the first matrix must equal the number of rows in the second. The outer numbers give the shape of the answer.

- a) Complete the table. Write “not defined” when a product cannot be computed.

Shape of A	Shape of B	Shape of AB	Shape of BA
2×3	3×4		
2×3	3×2		
2×2	2×2		

- b) Compute the product below.

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & -1 \\ -1 & 4 \end{bmatrix}.$$

- c) Let $H = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ stretch horizontally and $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ rotate counterclockwise by 90° . Compute RH and HR . Are they equal?

- d) Start with $\vec{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Compute $H\vec{v}$, then $R(H\vec{v})$. Check your result by computing $(RH)\vec{v}$. Which matrix acts first in RH ?